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Recent Trends and Techniques of Blood Glucose Level Prediction for Diabetes Control

July 2021

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Abstract

Diabetes, a metabolic disorder disease, can cause short-term acute or even long-term chronic complications in patients' body. In 2019, 9.3% of the world's adult population had diabetes, and these numbers are increasing day by day, which results in causes associated increase of morbidity, mortality, and health care cost related to diabetes. Thus, a huge research effort has been carried out to manage diabetes. A precursor to diabetes management is to predict the future blood glucose levels. In this paper, we provide a comprehensive and systematic study of diabetes management, focusing on recent research towards blood glucose level prediction. In particular, we have presented and categorized existing recent research based on major clinical application domains, different input features, and all major modeling techniques including physiological, data-driven, and hybrid models. We have summarized the performance analysis of different modeling techniques using different metrics, and critically analyze these techniques from different perspective. Finally, we have identified a number of research challenges that ranges from data collection to model improvement for T2DM type diabetes. This review can be a good starting point for researchers and practitioners who are working in building data-driven computational models for diabetes management and blood glucose level.

Keywords **-** Blood Glucose Level Prediction, Diabetes Management

1 Introduction

Diabetes Mellitus (DM) is a group of chronic metabolic disorders caused due to a raised level of blood glucose (BG) in a human body [38]. Diabetic patients may develop acute complications such as *ketoacidosis* that occurs when the body produces a high level of ketones, and *hyperosmolar* that occurs when blood has an extremely high sugar level, or chronic complications from long-term diabetes. Long-term diabetes may alter tissues and organs, which may result in diseases such as cardiovascular disease (heart disease), diabetic retinopathy (eye disease), diabetic nephropathy (kidney disease), and diabetic neuropathy (nerve disease). In recent years, the number of people with diabetes has increased at a staggering speed which causes an increase in morbidity, mortality, and health care cost. According to International Diabetes Federation (IDF) Diabetes Atlas 2019 [95], only in 2019, there were about 463 million adults aged between 20-79 years (9.3% of all adults in this age group) have diabetes.

Among them, 4.2 million adults were to die due to diabetes in 2019. This has triggered a considerable research effort to develop efficient approaches for diabetes management.

A large body of research focus on developing various diabetes clinical applications. These clinical applications include diabetes diagnosis, adverse glycemic events detection, diagnosis of diabetes-related long-term complications, Blood Glucose Level (BGL) prediction, Blood Glucose (BG) control, daily decision support system, prediction of population risk stratification, diabetes related sensor technology, identification of bio-markers, etc. We now briefly discuss some of them. Some research works diagnose diabetes or pre-diabetes using various input features such as Electronic Health Records (EHRs) [53, 67], Breath Signal Acetone Levels (BSALs) [112], and Heart Rate Variability (HRV). The works that predict adverse glycemic events i.e., hyperglycemia and hypoglycemia events include [79, 92, 105, 108, 114]. These applications use a wide variety of input features including historical BGLs, insulin dosage, meal intake, physical activities, and EHRs from hospitals or clinics.

Another group of research works in diabetes management focus on predicting long-term diabetic related complications including Diabetic Retinopathy (DR) [91, 107], Diabetic Neuropathy [50, 138], Diabetic Foot (DF) infection [69, 135], Diabetic Nephropathy (DN) [44, 96], and Cardio Vascular Diseases (CVDs) [45]. A major group of research works in diabetes management focus on building models to predict BGL. These modeling techniques include physiological models [4, 40, 97, 102], data-driven models [59, 78, 82, 83, 94, 119, 120, 123, 124], and hybrid models [28, 31, 33, 126, 134]. With the popularity and success of Deep Learning (DL) techniques, we have witnessed an explosive growth and the usage of data-driven models in this domain recently. Since there have been a large number of research works in many sub-fields of diabetic management, it demands a systemic review of related works in this domain.

There have been a few literature reviews on diabetes related clinical applications. Oviedo et al. [54] conducted a review on modeling strategies for personalized prediction of BGLs and adverse BG events. The reviewed modeling strategies include *physiological models*, *data-driven models*, *hybrid models*, and *control-relevant models*, which are published between 2010 and 2016. Kavakiotis et al. [64] conducted a systematic review of applications of only data-driven techniques on diabetes prediction, diabetic complications, biomarkers, and health care management systems. In another review, Dankwa-Mullan et al. [76] covered application of data-driven models in the Decision Support System (DSS), diabetes related complications, and prediction of population risk stratification. Contreras et al. [74] reviewed research works that proposed data-driven techniques between 2010 and 2018 in seven diabetes management (DM) categories including BG control, BGL Prediction, detection of adverse glycemic events, patient risk prediction, and DSS. Woldaregay et al. [115] reviewed the data-driven modeling in BGL Prediction related to Artificial Pancreas (AP) system, DSS, and adverse glycemic event detection. They reviewed 55 articles between 2010 and 2018 covering four application areas. Zhu et al. [141] reviewed 40 articles investigating the applications of DL techniques on diagnosis of diabetes, glucose management, and diagnosis of diabetes-related complications. Vettoretti et al. [136] provided a narrative review of the use of data-driven techniques for the development of DSS for BG control and BGL prediction.

Though the above reviews covered many sub-topics from different dimensions, no existing review work focuses on a comprehensive approach of BGL prediction that include input features, modeling techniques, performance comparisons, etc. To fill up the above research gaps, we present a systemic review of recent methods in BGL prediction. In summary, our major contributions in this paper are listed as follows.

- We have presented and categorized major clinical application domains in diabetes management, focusing on BGL prediction based application.
- We have organized related works of BGL prediction under two broad perspectives as follows: 1) input features, and 2) modeling techniques.
- As input features play a major role in the outcome of the BGL prediction, we have categorized existing research based on different types of input features for modeling.
- We have conducted an extensive review on recent techniques of all major modeling techniques

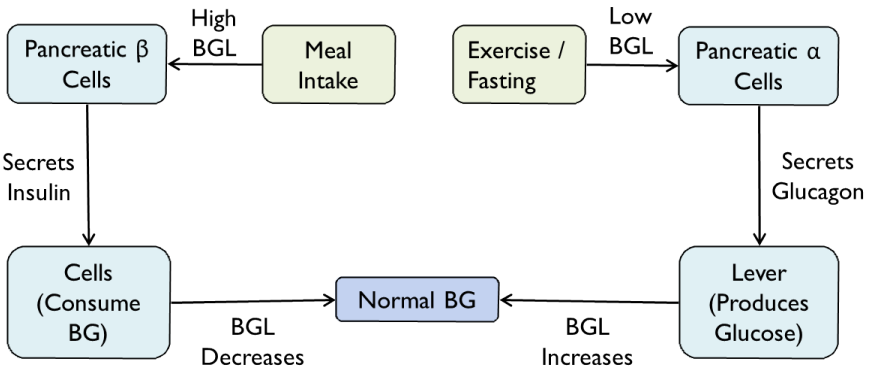


Figure 1: An overview of Glucose-Insulin dynamics

including physiological, data-driven, and hybrid models that have been used for the BGL prediction application.

- We have presented the performance of major works in BGL prediction using different metrics.
- We have identified a number of research challenges and the scope of future works, in particular, in the direction of BGL prediction of Type 2 Diabetes Mellitus (T2DM).

The rest of the paper is organized as follows. Section 2 provides some preliminary information on diabetes and highlights various clinical applications related to diabetes. In Section 3, the methodology of this survey is described. In Section 4, the role of different input features for BGL prediction is discussed. Section 5 discusses different modeling techniques for BGL prediction. Section 5.5 provides the comparative study of the performance measures and issues of different modeling techniques. Finally, Section 6 provides the conclusion and the open issues.

2 Diabetes: Background and Clinical Applications

In this section, we first briefly discuss the preliminaries about the physiological background of diabetes, and then discuss various clinical applications of diabetes.

2.1 Diabetes Basics

The increased BGL that occurs in DM is due to i) lack of secretion of insulin from pancreatic beta cell or ii) insulin resistance of cells, where cells fail to use insulin to consume glucose from blood or both.

Glucose Insulin Dynamics: The glucose-insulin regulatory system in a healthy human body as seen in Figure 1 provides the following impacts - (i) if there is high glucose in blood due to meal intake, the incretin hormones is secreted in the gut (a) to signal pancreas beta cell to release (bolus) insulin to match the glucose in the meal, (b) to signal pancreatic alpha cells to inhibit glucagon secretion to stop hepatic glucose production, (c) to slow gastric emptying to slow down postprandial glucose production. (ii) If there is high glucose in blood due to constant hepatic glucose production in the liver, pancreatic beta cells release constant drip of (basal) insulin to match that glucose. (iii) Insulin secreted from pancreatic beta cells allow the body cells to consume glucose from blood thus reducing glucose in the blood. (iv) If there is low glucose in blood due to physical exercise or fasting, pancreatic alpha cells release glucagon to stimulate hepatic glucose production breaking the glycogen in the liver. Besides that, kidney also contributes to the level of BG by determining the amount of glucose that can be reabsorbed by the blood and the amount that will be excreted through urine.

Types of Diabetes: The major two types of diabetes out of the various categories of diabetes according to the underlying cause of hyperglycemia listed by American Diabetes Association (ADA) [93] are - *Type* 1 diabetes and *Type* 2 diabetes. *Type* 1 diabetes is caused by an autoimmune process of a person born with genetic predisposition to getting the disease initiated by some environmental

trigger. This targets and starts destroying the pancreatic beta cells that produce insulin as foreign element. This leads to a progressive decline of pancreatic insulin secretion and eventually to the zero production of insulin. This type of diabetes can be developed at any age, cannot be prevented or reversed, and needs insulin treatment for rest of the person's life. Approximately, from 5 to 10% of all diabetic people have Type 1 diabetes [93]. *Type* 2 diabetes is caused by a pancreatic beta cell insufficiency that becomes unable to produce enough insulin or peripheral insulin resistance that reduces the ability of the cells to consume glucose from blood, or both. Obesity and the lack of exercise are two major factors to the development of Type 2 diabetes. Type 2 diabetes usually starts at late adulthood. However, it can be managed and controlled with healthy diet, physical activity, and/or medication. Sometimes, it even can be remitted.

2.2 Diabetes Related Clinical Applications

To prevent the long-term diabetes related complications, one has to *proactively* manage diabetes. Among many things, diabetes management plan should include - i) self-monitoring of BGL, ii) a meal plan to manage weight and BGL, iii) a physical exercise plan to lower BGL and increase sensitivity to medication, and iv) diabetes medication plan in form of oral dose or insulin injection to lower BGL. Diabetes engages two things: (a) a *biological process* that includes hormone action and metabolism, and (b) individual *human behavior* that is driven by personal preferences. On one side, every human being has unique biological process. In another side, every human has his/her own unique personal preferences. Due to these two unique things, every individual has a unique case of diabetes. Hence, each individual needs a separate diabetes management plan. People with diabetes with uncontrolled glucose in blood are at a higher risk of having life threatening complications in the long run.

Diabetes management starts with the diagnosis of diabetes onset. After the diabetes onset is diagnosed, the next step is to control the BGL. If the BG is not managed well, various diabetes related complications may arise. Those complications need to be predicted before they happen so that necessary steps can be taken to avoid them or push them further into the future. Hence, the diabetes related clinical applications can be broadly divided into two domains which are - *diagnosis oriented*, and *blood glucose oriented*. The categorized clinical applications are provided in Figure 2 and it is described in short as follows.

2.2.1 Diagnosis Oriented Applications

The diagnosis oriented clinical applications can be broadly divided into two domains which are - *diagnosis of onset*, and *diagnosis of complications*. They are described briefly as follows.

Diagnosis of Diabetes Onset: Early diagnosis of diabetes or *pre-diabetes* is very important. If

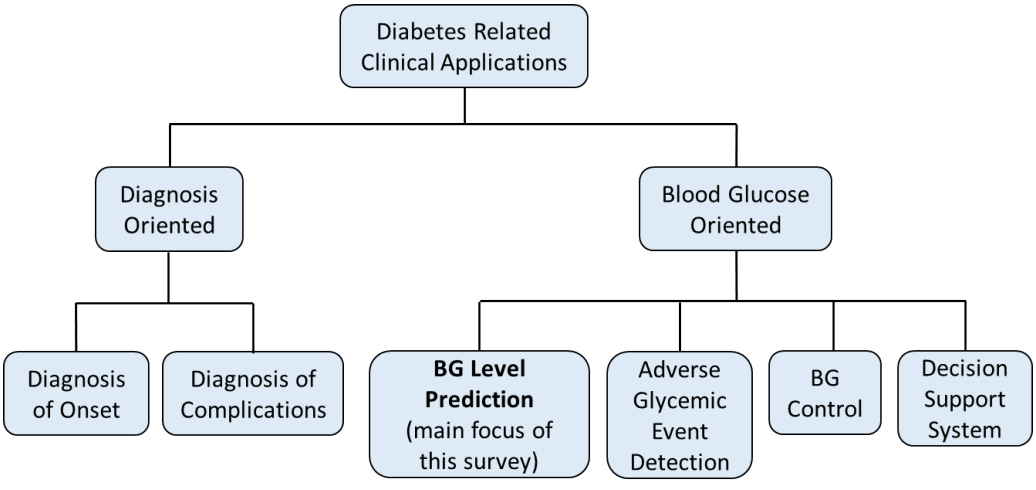


Figure 2: A taxonomy of clinical applications related to diabetes. (This taxonomy narrows search to BGL prediction.)

pre-diabetes can be diagnosed, appropriate actions e.g., change in diet, increase of physical exercises, etc. can be taken to avoid diabetes. If diabetes can be diagnosed at the early stage, change in lifestyle related to diet and physical activities can control the diabetes. It ultimately helps an individual to lead a normal life and to prevent diabetes related long-term complications. There are active researches in this field where to diagnose diabetes [53] used a 3-layer stack of de-noising autoencoders on aggregated EHRs, Pham et al. [67] introduced end-to-end prediction model *DeepCare* that used Long Short Term Memory (LSTM) on discrete EHRs, Srinivasan et al. [112] used 1-D Convolutional Neural Network (CNN) on real-time breath signals with acetone levels obtained from an array of gas sensors, Goutham et al. [79] presented a DL framework using LSTM and CNN on HRV signals, Ashiquzzaman et al. [58] presented a Deep Neural Network (Deep NN), Spanig et al. [111] employed Deep NNs using only demographic and basic physiological (non-invasive) data, etc.

Diagnosis of Diabetes-related Long-term Complications: Prolonged untreated elevated BGL, i.e., hyperglycemia, causes long-term diabetes-related complications [32]. Prolonged exposure of tissues (cells) to high concentration of glucose can lead to a severe damage of interior layer of blood vessels. This results in alteration of tissues and organs causing complications including DN, DR, neuropathy, or peripheral neuropathy. High glucose level can also damage big blood vessels. This may result in cardiovascular diseases and stroke. Due to peripheral neuropathy, one loses the ability to sense traumatic pressures on foot. This wears away the epidermis and exposes the deeper layers creating the neuropathic ulcer, then foot infection, and eventually turning to amputation.

In the field of DR, researchers used DL models [91] to predict future DR progression using Color Fundus Photographs of DR patients, used fundus photographs and a DL based system [113] as grading tool to determine the prevalence of the systemic cardiovascular risk factors for DR, used a DL algorithm [107] on gradable retina images to detect DR with specialist-level accuracy, used CNNs coupled with transfer learning and hyper-parameter tuning [86] to classify a given set of fundus images. In the field of diabetic neuropathy, researchers developed a CNN model with data augmentation [138] for the diagnosis of diabetic neuropathy, identified variables that are correlated with a diagnosis of Diabetic Peripheral Neuropathy [50] using RF modeling applied to EHR, used a hybrid approach employing both Machine Learning (ML) and rule-based clinical text mining techniques [45] to extract related information on CVD risk factors from unstructured clinical notes.

In the field of DF, researchers analyzed the use of AI and DL techniques [135] for the classification of DF thermograms, and presented a CNN architecture, DFUNet [62] to diagnose DF Ulcers (DFU) using foot images. Fiarni et al. [96] constructed a prediction model for retinopathy, nephropathy, and neuropathy using Naïve Bayes Tree and C4.5 decision tree-based classification techniques, and k-means clustering techniques, and found the correlated feature and sub feature as a disease risk factor.

2.2.2 Blood Glucose Oriented Applications

When diabetes is already diagnosed, lifestyle must be changed to reduce the BG. This ultimately helps a patient to avoid further long-term health hazards. Healthy lifestyle might even cure a patient from diabetes. Therefore, it is very important to explore the possible steps for the BG management domain which is full of diversified challenges and issues. This BG Management research domain can be broadly divided into four sub-domains: BGL prediction, adverse glycemic event detection, BG control, and the daily decision support system. They are described in short as follows.

BGL Prediction: The basic goal of a glucose management plan is to keep the BGL within the normal range, which is between 90 mg/dl and 130 mg/dl before meals, and under 180 mg/dl two hours after the meals. BGL prediction has goals (i) to prevent too low BGL, which is called *hypoglycemia*, (ii) to prevent too high BGL, which is called *hyperglycemia*, or (iii) to prevent or delay long term diabetes related complications, such as *DR*, *neuropathy*, CVDs like *heart attack* or stroke, DN, etc. The previous researches on BGL prediction models have been extensively covered in Section 5.

Adverse Glycemic Event Detection: A healthy person's BG remains in the normal level. This is controlled by the insulin secreted from the pancreatic beta cell. For a diabetic person, due to one or multiple of the following problems: autoimmune system disorder, cell glucose resistance, and reduced insulin secretion capacity of pancreatic beta cell, normal BGL cannot be maintained. For many diabetic persons, the BGL goes over the normal level, which is termed as *hyper-glycaemia*. Various

complications occur due to hyper-glycaemia including CVDs, DR, DN, neuropathy, DF infection, etc. On the other hand, for many the BGL go below the normal level, which is termed as *hypo-glycaemia*. The complications that might occur due to prolonged hypo-glycaemia are seizure, unconsciousness, and in extreme cases even death. There are active researches in this field where [92] employed an Extreme Gradient-Boosted (XGB) tree algorithm to classify the post-prandial glycemic status, Oviedo et al. [105] proposed bi-class Support Vector Classifier for predicting postprandial hypoglycemia, Reddy et al. [108] used Random Forest (RF) classifier to avoid exercise related hypoglycemia, Vehi et al. [114] proposed a Support Vector Machines (SVMs) with Radial Basis Function (RBF) kernel based model to predict hypoglycemic events during postprandial periods, and Artificial Neural Network (ANN) to predict hypoglycemic episodes overnight, etc.

BG Control: High level of sugar in blood causes long term complications. To prevent complications, a person needs to bring the high blood sugar level down within the normal range. To lower the BG, one can take diabetes medication in the form of oral dose or insulin, can do a rigorous exercise for a while, bring a change in his/her diet, or any combination of them. There are active researches in this field where [46] stress-tested the performance of a dose safety fuzzy logic controller [21, 22] to highlight the deficits in the AP system, Daskalaki et al. [49] presented an RL-based algorithm for personalized insulin infusion and glucose regulation in T1DM, Favero et al. [41] used an improved version of Modular Model Predictive Control [26] to control a Continuous Subcutaneous Insulin Infusion (CSII) in a multi-center outpatient setting, Cappon et al. [71] developed an NN corrector consisting of a fully connected Feed-forward NN (FNN) with 3 hidden layers, Seo et al. [110] developed algorithms based on a RF, a SVM using a linear function or a RBF, a K-Nearest Neighbor (KNN), and a logistic regression to control postprandial hypoglycemia, etc.

Daily Decision Support System (DSS): Decision support systems are open loop systems that provide recommendations to a diabetic patient to manage diabetes. These systems provides the appropriate insulin dosage amount to ingest, sometimes along with the BGL prediction. Sometimes, it might recommend to take a 30 minute brisk walk, or to talk to the doctor. There are active researches in this field where [68] proposed a Reinforcement Learning (RL) algorithm to improve participants' compliance with the activity program, Sun et al. [84] proposed an adaptive basal-bolus algorithm based on RL to provide personalized adaptive insulin optimization and achieve glucose control, Torrent et al. [85] proposed PepperRec, a Case-Based Reasoning (CBR) bolus insulin recommender system to keep BGL in the target glycemic range longer, etc. The diabetes related clinical applications and associated research works are given in Table 1.

3 Overview of the Survey

3.1 Major Focus

The field of diabetes has a vibrant research community working on various diabetes related clinical applications. Out of them BGL prediction is the most popular field among the researchers. This is due to its use in other fields such as BG control, prediction of adverse glycemic events, daily support system, etc. A predicted BGL value will also empower a diabetic patient to self manage his/her BGL. The focus of this survey is to present a comprehensive review of recent modeling techniques in BGL prediction.

3.2 Methodology

An extensive literature search has been conducted on three online databases PubMed, Google Scholar, and ResearchGate. The search was restricted to English language peer reviewed high impact journals, articles, and conference proceedings published between 2015 and June 2021. The search keywords were blood glucose level or dynamics or concentration, prediction or forecasting, and diabetes. According to the focus of this survey, all types of modeling techniques for BGL prediction both for Type 1 and Type 2 diabetes were included. Finally, after applying all the above inclusion and exclusion criteria 58 papers were included in this review as shown in Table 4 and Table 5.

Table 1: Categorized clinical applications - Diabetic Retinopathy (DR), Diabetic Neuropathy, Cardio Vascular Disease (CVD), Diabetic Foot (DF) infection, Diabetic Nephropathy (DN)

Application		Ref.	Description
Diagnosis Oriented Applications	Diagnosis of Diabetes Onset	[53, 58, 67, 79, 111, 112]	Diagnose diabetes, prediabetes, etc.
	Diagnosis of Diabetes-related Long-term Complications	DR - [86, 91, 107, 113], Neuropathy - [50, 138], CVD - [45], DF - [61, 62, 69, 135], DN - [44, 96]	Diagnose diabetes-related long-term complications.
Blood Glucose Oriented Applications	BGL Prediction	Given in Table 4 and 5 in detail	Predicts BGL.
	Adverse Glycemic Event Detection	[92, 105, 108, 114]	Predicts hyperglycemia and hypoglycemia events.
	BG Control	[21, 22, 26, 41, 46, 49, 56, 71, 110]	Find solution e.g., proper insulin dosage amount to control normal BGL. Automatic Pancreatic (AP) System
	Daily DSS	[68, 75, 77, 84, 85]	Provide daily recommendations to manage diabetes.

A typical BGL prediction framework is given in Figure 3. Both selection of contributing input features and design of efficient modeling techniques play vital part in BGL prediction. We provided categorization of the reviewed research works based on input features and modeling techniques as shown in Figure 4. In case of input features, the reviewed articles were categorized based on the most widely used input features: previous readings of BGL, insulin dosage amount, physical activity, and meal intake. A detailed study on this categorization is presented in Section 4. In case of modeling techniques, the reviewed articles were categorized based on all currently used modeling techniques: physiological models, data driven models, and hybrid models. A detailed study on this categorization is presented in Section 5.



Figure 3: A typical BGL prediction framework

4 Input Data Used in Different Models

There are quite a few factors that influence the BGL. To better predict the BGL, enough data on these factors along with the previous readings of BGL are used as an input to the prediction system. The mostly used input feature is the pre-recorded *BGLs*. 96% of the reviewed research articles mentioned this as the input features. The second most used input feature is the history of *meal intake*. 52% of the reviewed research articles used some forms of meal intake as input feature. The third most used input feature is the history of *insulin dosage*. 50% of the reviewed articles included this as input feature. The fourth most used input feature is the history of *physical activities*. 28% of the reviewed articles included this as input features. Figure 5 presents the taxonomy of input features. A detailed study on these input features is presented below.

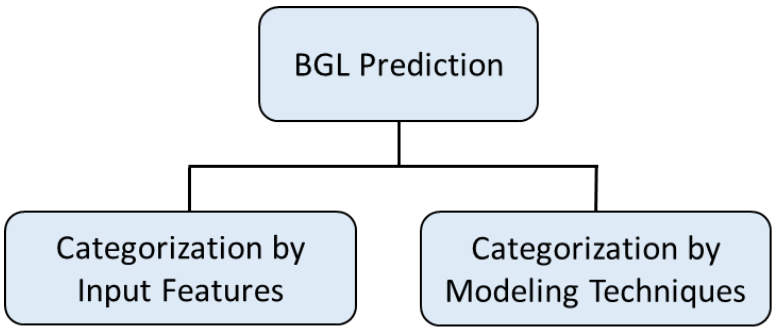


Figure 4: A taxonomy based on the way research works are categorized

4.1 Blood Glucose Level

BGL is one of the most important factors that influence the prediction of BGL. An individual who has Type 1 Diabetes Mellitus (T1DM), must use a Continuous Glucose Monitoring (CGM) device to monitor his BGL. A CGM device used in [36, 57, 66, 78, 100, 102, 106, 117, 121, 123, 124, 126, 127] usually provides a BGL reading every five (5) minutes. That means, it can provide 288 readings per day, which is a lot of data points per day. For a T1DM patient, in most of the cases, 30 days of CGM data is enough to predict BGL.

On the other hand, an individual who has Type 2 Diabetes Mellitus (T2DM), does not need to measure continuous BG reading. Therefore, the patient may use a hand-held blood glucose meter used in [94, 128] to measure own blood-glucose. Usually, the patient can measure before and after two hours of each meal, which amounts to 6 data points per day. That being the case, data collection for T2DM patients is more challenging than T1DM patients.

As CGM data from T1DM participants can be collected with less efforts, majority of the research in BGL prediction focus on CGM based prediction. 83% of the reviewed articles collected data from CGM devices. Whereas only 2% of the reviewed articles used reading from hand-held blood glucose meters from T2DM patients. 9% of the articles used both. Few research works did not provide any information on the data collection process.

4.2 Insulin Dosage

A meal ingestion at any time increases the BGL of a diabetic patient. Due to insulin secretion deficiency, a diabetic person needs to take insulin in the form of oral medication or exogenous insulin ingestion subcutaneously using subcutaneous insulin syringes or insulin pens [31, 42, 97, 128], or CSII using insulin pump [83, 100, 104, 106, 116, 119, 128, 130, 139]. The insulin dosage allows body cells to consume glucose from blood. This reduces the glucose level in the blood [1]. However, in most of the clinical cases the dosage and time for each insulin ingestion are manually recorded by the diabetic

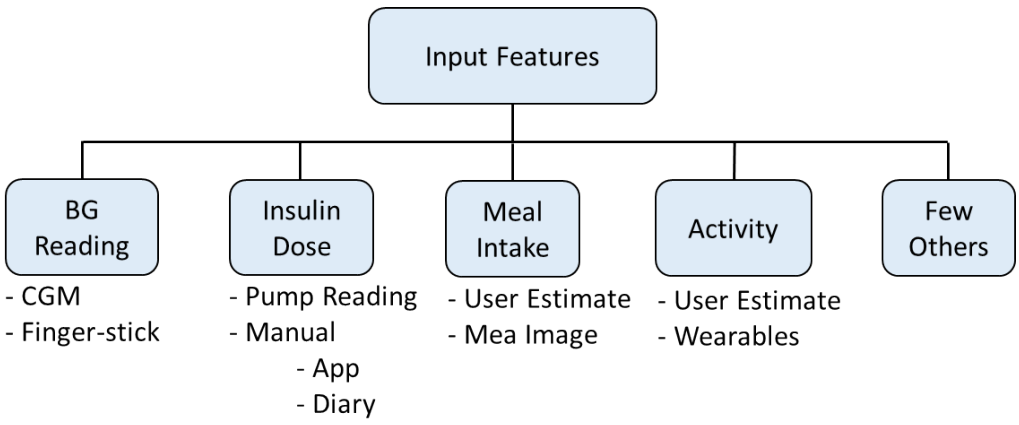


Figure 5: A taxonomy based on the input features of models

patient using paper-diary or smart phone apps. In case of pump, users need to enter the amount of insulin in the insulin pump. Some insulin pumps can store the dosage amount and time for few months.

4.3 Physical Activity

Physical activity or exercise is one of most important factors that influences BGL. Physical activity triggers consumption of glucose from blood by body's muscle cells, which results in reduction of BGL [9, 17, 37]. Only 28% of the reviewed articles used physical activity as input features. Among these cases, 69% used [34, 42, 104, 119, 121, 124, 139, 141] some type of wearable devices to count the step count. In few of the cases mentioned in [97, 100], participants manually entered the estimated numbers. The research works that used wearable devices, considered Heart Rate (HR), Galvanic Skin Response (GSR), skin temperature, and air temperature along with the step count. A few percentage of the research works did not mention the method of recording the physical activity.

4.4 Meal Intake

Meal intake is the primary source for the increase of glucose in the blood. A meal can have carbohydrate (CHO), protein, fat, fiber, and vitamins & minerals. Protein, fat, fiber, and vitamins & minerals do not contribute to the increase of glucose in blood. The CHO of meal is the sole source of increase of glucose level in blood [13]. To predict BGL accurately, the CHO content (in grams) of each meal must be computed. For an individual to self-manage his/her diabetes, he/she needs to estimate the CHO content (in grams) of each meal. In 75% of the research articles, the collected CHO amounts are directly estimated by the diabetic persons themselves either in the amount of grams or meal type and meal size. All the different ways of handling meal intakes used in the reviewed articles are discussed in detail in the following subsections.

4.4.1 Patient Estimated CHO Amount

Patients recorded the estimated CHO amount in grams using several ways including smart phone app [94, 100, 104, 116, 128, 130, 139], or manually recorded in a diary [36], or special applications other than common smart device application. It was not mentioned how patients estimated the CHO amount. There were few papers which did not mention the mode of recording the CHO amount [52, 106].

4.4.2 Patient Estimated – Dietician Annotated

Georga et al. [34, 42] collected data from fifteen Type 1 diabetic patients in free living condition. The patients recorded the meal intake, i.e., type of food, serving size, and time themselves. Later on, a dietician analyzed the meal and provided the estimated CHO amount. However, this data collection is a bit lengthy procedure, and it needs the approval of a Dietician.

4.4.3 Patient Estimated – System Computed Macronutrient Amount

Karim et al. [99, 126] carried out the clinical studies at the Cardiac Rehabilitation Institute of the Military Hospital, Balatonfüred, Hungary. The patients had daily activities that were similar to free living condition. In 2015 [99], they collected detailed dietary log of 21 days of 16 T2DM volunteer patients. In 2019 [126], they collected detailed dietary log of 84 days of five volunteer patients of both Type 1 and Type 2 diabetes. The patients logged meal information using a lifestyle support Lavinia Lifestyle Mirror smart phone application and the MenuGene dietary expert database to find the lipid, protein, dietary fiber, monosaccharide, starch quantity, and glycemic index (GI) values of every logged meal. Gyuk et al. [97] were also a part of the same study that was carried out in 2015 at the Cardiac Rehabilitation Institute of the Military Hospital above except that they collected data of 26 T2DM patients.

4.4.4 Blind Estimation of Meal Absorption

Liu et al. [102] carried out their clinical trial at Imperial College Healthcare NHS Trust, London, UK, on ten T1DM adults, for two weeks. Due to the unavailability of any reliable information on meal absorption, they assumed that breakfast and snacks were fast absorption meals, and lunch and dinner were medium absorption meals.

4.4.5 CHO Estimation from Food Image

6.67% of the reviewed articles that have used meal as one of the input features used food image as the initial source of meal consumed. The data used by [36] was collected during the European Union Seventh Framework Program (FP7) DIAdvisor project. In that project, as mentioned by the authors, the information on ingested CHO was extracted from the picture of meals taken by the patients with a standard mobile phone with camera. Though what was done with the pictures was not mentioned.

Chakrabarty et al. [72] used a publicly available food dataset Food-101 to train their Deep NN model. This Deep NN model predicts the food category with a confidence value. They tested this model using the UVA/Padova metabolic simulator. When meal is announced in the simulator, a unique image from Food-101 test set was randomly selected. The Deep NN model then predicted the food category. If the confidence value was above some threshold, they found the macro-nutrient content of the food using a USDA nutritional database. Then they computed the amount of bolus insulin dosage.

Saiti et al. [134] used two sources of data - one in collaboration with FN Motol University Hospital for training purpose, and for the second they used the open D1namo dataset. Both datasets included food pictures. One used the consumed CHO estimated by the physicians, and the other used estimated amount of calories, quality, and whether balanced or not by dieticians. Munoz et al. [132] also used the open D1namo dataset.

4.5 Other Features

Besides those four critical factors researchers used other input features as well. Dave et al. [123] used demographic features age, gender, BMI, HbA1c, duration, and few derived features of CGM readings along with BGL, insulin and meal intake. Faruqui et al. [94] besides BGL, exercise and meal intake, used weight. Gyuk et al. [97] besides the critical four factors, used medication logs to predict BGL.

Xie et al. [139] besides using the main four factors used all other data that were available in the OhioT1DM dataset that included data from wearables (HR, GSR, skin temperature, and air temperature), sleep, work, stress and illness. Kushner et al. [128] besides BGL, insulin, meal, and step count, used demographic values: age, weight, duration, heart rate, and HbA1c. Along with BGL, insulin, and meal intake, weight of the patients were used by [101] to forecast accurate BGL.

Table 2: Category of different input features

Features	Recording Method	Ref.	Description
Meal	Patient Estimated CHO Amount	[36, 94, 100, 104, 116, 128, 130, 139]	Patients estimated the CHO amount in gram of their meals.
	Patient Estimated – Dietician Annotated	[34, 42]	Patient recorded meal type and size, dietician estimated CHO amount.
	Patient Estimated – System Computed	[97, 99, 126]	Patient recorded meal type and size, system computed amount of CHO, protein, and lipid.
	Blind Estimation of Meal Absorption	[102]	Assumed the meal absorption rate: Fast, Medium, Slow.
	CHO Estimation from Food Image	[36, 72, 132, 134]	CHO amount is computed from meal pictures taken by patients.

Continued on next page

Table 2 – continued from previous page

Features	Recording Method	Ref.	Description
BG Reading	CGM	[36, 57, 66, 78, 100, 102, 106, 117, 121, 123, 124, 126, 127]	Used CGM devices to record BGL.
	Finger tip BGL	[99, 119, 128, 130, 132, 139, 141]	Used ordinary fingertip blood glucose meters.
	FGM	[121]	Used a flash glucose meter sensor under skin.
Insulin dosage	Insulin Pump Reading	[83, 100, 104, 106, 116, 119, 128, 130, 139]	Used insulin pumps to apply insulin.
	Smart Phone App	[94, 100, 104, 116, 128, 130, 139]	Recorded insulin dosages using a smartphone app.
	Diary Entry	[31, 42]	Record insulin dosages using a paper diary.
Physical Activity	Wearables	[34, 42, 104, 119, 121, 124, 139, 141]	Used fit band like wearables to record physical activity (step counts).
	Patient Estimate	[97, 100]	Patients recorded estimated physical activity manually.

4.6 Critical Analysis

As high volume of CGM data can be collected from T1DM patients, most of the existing research focus on developing BGL prediction models from CGM data. However, the proposed models that used T1DM patient data cannot be readily used to predict BGL for T2DM patients. More efforts should be given to collect data from T2DM patients. Insulin dosage recordings were mostly taken by the patients manually either by entering the amount into insulin pump, or recording using a smart phone app, or a paper diary. The use of insulin pump is probably overkill for the T2DM patients. A smart phone application based insulin recordings can be a feasible solution for collecting insulin intake of T2DM patients. To capture the physical activity related data of the patients, wearables such as *Fitbit* wrist bands are being used by the researchers for data input. More research needs to be aimed to identify the type of activities and intensity of the activities.

Meal consumption data are mostly recorded manually by the patients, where patients estimated the CHO amount of their meals. Sometimes, the dieticians annotated the CHO amount of the meals from meal pictures taken by the patients. However, these are either prone to be highly inaccurate or highly dependent on the knowledge of the experts. Predicting the CHO amount from the meal picture is an interesting and promising area of research. It removes the burden of manual intervention of estimating CHO amount from the patients. However, only a few papers used meal images to record the meal intake of the diabetic patient. More researches need to be focused on capturing image of meal, predicting CHO amount, and finally predicting the BGL of the patients. The input features and the ways they have been recorded are given in Table 2.

5 Modeling Techniques

In this section, we discuss the proposed modeling techniques for BGL prediction. Figure 6 presents the taxonomy of modeling techniques. The comparative study of performance of those modeling techniques is analyzed in Section 5.5.

BGL prediction is a challenging task as BGL depends upon may factors. Thus, BGL prediction models take different types of input data including daily amount of insulin ingestion, amount of meal intake, duration of physical exercise and the previous history of BGLs. Besides these four

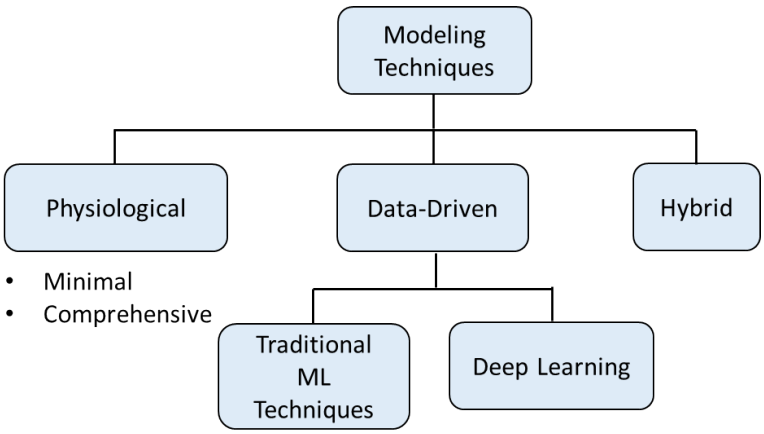


Figure 6: An overview of different modeling techniques

main BG movers, demographic data, HbA1c, sleep, stress, disease history, diabetes duration, smoking history, alcohol consumption history, etc. are also used. The volume of the input data also plays a key role in the BGL prediction, i.e., the more data is used, the more accurate the BGL prediction can be achieved. BGL prediction models also need to consider insulin absorption dynamics, insulin effects on glucose appearance rates and disposal rates, meal consumption dynamics, and variability in physiology of patients. Many research efforts have been focused on finding BGL prediction models. These BGL predictive models can be classified into the following three main categories depending on the computational methods they used (i) *Physiological* models (PM), (ii) *Data Driven* models, and (iii) *Hybrid* models (HM). The categorized modeling techniques are given in Table 3 and described in detail as follows.

5.1 Physiological Models

Physiological models use the actual knowledge of physiology to capture the glucose-insulin dynamics that is described in Section 2.1. The major organs of the human body that are involved in the glucose-insulin dynamics include intestines, brain, lever, heart, kidney, pancreas, brain, lungs, vascular organs, peripheral tissues, etc. These models use multi-compartmental models including carbohydrate digestion model, glucose absorption model, insulin absorption model, insulin action model, etc. to mathematically represent some or all the above mentioned organs. The variations in these models depend on the underlying physiological aspects and the number of organs represented by the used compartments. Based on these two factors, the physiological models can further be categorized into (i) minimal models and (ii) comprehensive models [24].

5.1.1 Minimal Models

In *minimal models*, various organs and cells are ignored due to their minimal contribution to the glucose-insulin dynamics. These models used a few compartments to represent the organs that contribute the most but still achieving satisfactory results with computational efficiency. *Bergman's Minimal* model [4] is one of the widely used minimal models. It uses the plasma glucose, the plasma insulin, the insulin action on glucose, and the meal absorption compartments.

Liu et al. [102] used a compartmental composite model of glucose-insulin dynamics that used a deconvolution technique on CGM signals for estimating the state [29]. Additionally, they used the CHO amount, meal absorption information, and insulin boluses as exogenous inputs. Their result shows that inclusion of meal absorption information enhanced the prediction accuracy. Gyuk et al. [97] proposed an algorithm that combined two things. One is a two compartmental algorithm [11] for meal absorption. The other is an advanced glucose control system that used small set of parameters but still maintaining powerful descriptive capabilities based on Delay Differential Equations (DDE) [25]. They extended the combined model with personalized parameter training with Genetic Algorithm

Table 3: Category of different modeling techniques

Models	Category	Ref.	Description
Physiological	Minimal	[4, 40, 97, 102]	Various organs and cells are lumped into one or more compartments.
	Comprehensive	[6]	All the organs that are involved in the glucose and insulin dynamics are modeled separately.
Data Driven	Traditional ML	[34, 43, 51, 59, 80, 82, 87, 88, 89, 90, 117, 118, 120, 122, 123, 128, 129, 133]	Used traditional machine learning techniques for prediction.
	Deep Learning	[47, 65, 66, 73, 78, 83, 94, 100, 101, 104, 116, 119, 124, 127, 131, 132, 139, 140]	Used DL techniques for prediction.
Hybrid	Physiological + ML	[28, 31, 33, 126, 134]	Minimal model and Traditional ML techniques.
	Physiological + Deep Learning	[81, 130], etc.	Minimal + Deep Learning.
	Physiological + GE	[60]	Grammatical evolution takes physiological model as input.

(GA) and Nelder-Mead method and used a diurnal parameter profile as a representation of the natural biorhythm.

Bock et al. [40] proposed a new linear dynamic Therapy Parameter-based compartmental Model (TPM) which is based on the static prediction model of standard therapy and identifiable on BGL measurement only. The model parameters are intrinsically correlated with physician-set therapy parameters. The model equations of the TPM are derived from the Bergman Minimal model [4].

5.1.2 Comprehensive Models

In the comprehensive models, all the organs and the cells that are involved in the glucose and insulin dynamics are considered and modeled separately using separate compartments. Tiran et al. [2] firstly proposed a comprehensive physiological model. Guyton et al. [3] and Cobelli et al. [5] proposed other physiological comprehensive models. Sorensen et al. Sorensen et al. [6] provided a detailed mathematical explanation of the work of Guyton et al. Due to their high complexity, comprehensive models are not being used by the larger research community.

5.2 Data Driven Models

In this section, different aspects of data driven models for prediction of BGL are discussed. The data driven models use large volume of diabetic patients' related data to extract valuable patterns in the data. These models do not depend on underlying domain related expert knowledge i.e., human physiological information. Data driven models can be broadly divided into the following categories: traditional Machine Learning (ML) based models and Deep Learning (DL) based models.

5.2.1 Machine Learning (ML) Based Models

Dave et al. [123] used logistic regression with *Least Absolute Shrinkage and Selection Operator* (LASSO) and Random Forest (RF) with Variable Importance Plot (VIP) as two feature selection strategies to predict future glucose values where CGM values, insulin and CHO on board, patient demographic information and other CGM derived values were taken as input. They were not only able to accurately predict glucose values for lower PH, but they also found PH up to 60 minutes with more

than 90% accuracy. Bamgbose et al. [59] used a two-layer NN model with 8 hidden neurons as the modeling technique. They applied the NN model on AIDA simulated data that includes 23 features including glucose, insulin, and meal. Alqudah et al. [120] used ANN with one hidden layer, six neurons and seven inputs to find correlation between their input features (glucose, insulin, demographic, etc.) and BGL. Kushner et al. [128] utilized physiologically-informed feed-forward neural network (FNN) incorporating implicit biological facts by constraining training weights to match physiological knowledge. They partitioned the first layer into insulin and non-insulin inputs (glucose, meal, demographic, etc) to mimic learning "insulin on board". They used adaptive transfer learning in case of limited data.

Marcus et al. [129] used four patient specific supervised machine learning models - four Kernel Ridge Regression (KRR) variants: KRR and KRR RBF with the regular and special labels to compute BGL taking only CGM values as input. Alfian et al. [118] proposed Extreme Gradient Boosting (XGBoost) [48], an ensemble learning model that utilized gradient boosting, CGM values, other statistical-based features, and multi-step forecasting. Perez-Gandia et al. [82] used fully connected FNN with three layers, where first layer had 10 neurons, second layer had five neurons. All the layers had "sigmoidal" activation functions to predict BGL using only CGM values as input.

Georga et al. [43] investigated the possibility of applying Extreme Learning Machines (ELMs) to predict the glucose time series from CGM, insulin, exercise, and meal inputs. Frandes et al. [51] proposed a risk based non-linear Auto-Regressive Neural Network (AR-NN) based approach to forecast BG dynamics with an integrated measure of intra-patient BGL variability using CGM values only. DAntoni et al. [122] proposed an Auto Regressive Time Delayed (ARTiDe) Jump Neural Network (Jump NN) with feedback connections from output to hidden layer. They had time delays for each of the input-to-hidden, output-to-hidden, and input-to-output connections to predict BGL using only small amount of glucose data. Yang et al. [88] developed an ARIMA model with adaptive identification algorithm to model the time-varying non-stationarity of CGM data by adjusting the model orders and simultaneously updating model parameters based on the Akaike Information Criterion (AID) and Least Square Estimation (LSE) methods.

Yu et al. [89] proposed a data-driven prediction strategy that employed four adaptive filtering mechanisms: recursive least squares (RLS), extended recursive least squares (EX-RLS), kernel recursive least squares (KRLS), extended kernel recursive least squares (EX-KRLS) for training the candidate models using only the CGM values as input. Among the four candidate models, it then selected the prediction results of the model with lowest mean value of the prediction error, and the model with the lowest current learning error. It then fused the two selected prediction outputs with linear stack weighted method. Finally, it applied an additional smoothing filter to mitigate the noise and abrupt variability to predict the BGL. Yu et al. [90] developed adaptive kernel sparse filtering algorithms called *Approximate Linear Dependency Kernel Recursive Least-Squares* (ALD-KRLS), and *Surprise Criterion Kernel Recursive Least-Squares* (SC-KRLS) to provide real-time prediction of BGL using only CGM values as input.

Zarkogianni et al. [47] presented a comparative assessment of four glucose prediction models based on FNN, Self-Organizing Map (SOM), neuro-fuzzy network with wavelets as activation functions (WFNN), and linear regression model using glucose and physical exercise as input. Xie et al. [87] conducted comparative analysis of two DL approaches: a vanilla LSTM Network and a Temporal Convolution Network (TCN) and few other popular ML algorithms: the classic Auto Regression with Exogenous inputs (ARX) model, Huber Regression, Ridge Regression, Elastic Net Regression, Lasso Regression, Support Vector Regression (SVR) with Linear Kernel, Support Vector with Radial Basis Kernel, RF, and Gradient Boosting Trees for BGL prediction. To perform multi-step prediction they used recursive and direct methods. They used the OhioT1DM dataset (glucose, insulin, exercise, meal, heart rate, etc). In [139], Xie et al. added significant modifications and new contents by adding evaluation of the effect on BGL prediction with respect to different set of features.

Hamdi et al. [80] proposed DE-SVR, a SVR model based on Differential Evolution (DE) algorithms for prediction of continuous BG using only CGM values as input. Bertachi et al. [70] presented a model based on Artificial Neural Networks (ANN), where feature space had 5 features that came from CGM, Insulin Pump, and Fit band readings and physiological models. Ozogur et al. [133] proposed

average and patient specific modified fuzzy particle swarm optimization algorithm for the prediction of BGLs from CGM values. They used fuzzy Time Series for the partitioning of the search domain and Particle Swarm Optimization (PSO) for finding the optimal solution. [63] presented a set of evolutionary methods based on Genetic Programming (GP) and Grammatical Evolution (GE) and compared them with two ML techniques RF and KNN using history of BGL, insulin dosages, and meal as input.

Reymann et al. [55] proposed a SVR model for BGL prediction using only simulated BGL values as input.Mayo et al. [103] proposed a linear SVR model trained with normal and polynomial features for normo and hyperglycemic BGL prediction and a MLP trained on oversampled data using Adaptive Synthetic (ADASYN) for hypoglycemic BGL prediction to handle the imbalanced regression problem using BGL values. He et al. [125] used Kernel Canonical Correlation Analysis based on Particle Swarm Optimization (PSO-KCCA) with an error correction to predict BG concentration using BGL values. Prendin et al. [143] compared thirty different linear and nonlinear predictive algorithms using BGL values as input. Rodriguez et al. [109] developed and compared person-centric short-term prediction models based on ARIMA, RF, and SVM minimizing data variability, volume, and velocity of BGL values. Hobbs et al. [98] developed a BGL prediction model applying the Predictor-Based Subspace Identification (PBSID) including heart rate along with CHO intake, and insulin dosages.

5.2.2 Deep Learning Based Models

Deep Learning (DL) based techniques, which are basically ANNs with many hidden layers, are getting a lot of attention from the researchers recently in diabetes diagnosis and for predicting the BGL. Sun et al. Sun et al. [83] used a sequential model with one LSTM [10, 12] layer, one BiLSTM layer and several fully connected (FC) layers to predict BGL for different PHs using only CGM values. In their case, the LSTM method outperformed the baseline ARIMA method with lower RMSE values in all PHs. Faruqui et al[94] proposed two transfer learning strategies where glucose levels, exercise, meal intake, and weight were used as inputs. In one strategy, they used all patients' data to pre-train a global LSTM model. In another strategy, they found the similarity in glucose patterns between each pair of patients using Dynamic Time Warping (DTW). Then, they created a transfer learning dataset for each patient sampling from the similar patients to pre-train a global LSTM. Finally, for both of the strategies they used the global LSTM model to personalize each patient using individual patient's data.

Gu et al. [124] used a Neural Physiological Encoder (NPE), which is basically CNN applied on separate input features like CGM and CHO values, insulin dosage, and time with m kernel filters randomly initialized n times, then concatenated. The aggregate feature (the output of NPE) is being fed to an LSTM for BGL prediction. Kim et al. [127] used Gated Recurrent Unit (GRU) with a single RNN layer as the sequence processing layer and two stacked dense layers as regression layer to capture the patterns from the individual patients' time series of continuous glucose readings for glucose prediction. Aiello et al. [116] proposed a DL approach for forecasting glucose levels based on a pair of models composed of stacked LSTM networks. One model took past CGM, insulin, and CHO intakes as input and the other model received future insulin and CHO values. The output of both the models were concatenated and fed through Fully Connected (FC) layer for the prediction of future BGLs.

Mirshekarian et al. [66] proposed a Recursive Neural Network (RNN) approach that used LSTM units to learn physiological model of BG from CGM and CHO values and insulin dosages. Mirshekarian et al. [104] introduced a general memory-augmented MemLSTM neural attention model architecture, specially designed for time series prediction. They presented the results of a much larger set of experiments on real and synthetic data with the same features in [66] in what-if, agnostic, and inertial scenarios. Mohebbi et al. [131] investigated and compared the use of population-based and patient-specific LSTM based RNNs to patient-specific Autoregressive Integrated Moving Average (ARIMA) models using CGM values. Li et al. [100] proposed a multi-layer Convolutional Recurrent Neural Network (CRNN) architecture for glucose prediction where CGM values, insulin dosage, exercise, and CHO amounts are taken as input. The CRNN model was composed of three parts like, a multi-layer CNN that extracts the data features using convolution and pooling, followed by a RNN layer with

LSTM cells and FC layers. The CNN layer comprised a 1D Gaussian kernel filter to perform the temporal convolution. The pooling layers are used for reducing feature set. Li et al. [101] introduced GluNet, a personalized deep NN framework consisting of data preprocessing, label transform and recover, multi-layers of dilated CNN, and post-processing to predict the probabilistic distribution of short-term BGLs based on historical data as in [100].

Mhaskar et al. [65] proposed a semi-supervised parsimonious DL method using domain knowledge to suggest the compositional structure of the target function as well as the values of the constituent functions based on ideas in [35] using only CGM values. They first constructed three networks based on whether a 5-minute prediction indicates the trend to be in the *hypo-*, *eu-*, or *hyper-glycemic* range. Then, they trained a "judge" network to get a final prediction based on those three outputs. They used diffusion geometry ideas in [27] to train networks in manifold learning manner. Finally, they used a technique similar to the meta-learning approach in [30] where they used the entire data for 30% of the patients as training data, and predict the BGL for the remaining 70% of the patients in the data set. Fox et al. [78] used a sequential polynomial multi-output deep RNN (PolySeqMO) to predict BGL trajectories using only glucose levels.

Alfian et al. [117] proposed a Multi-layer Perceptron (MLP), an ANN as prediction model. They used direct method and sliding window strategy to convert the time series CGM data into a set of paired inputs and outputs. Additional statistical features were then added to these features and fed to the MLP model. They used the grid search algorithm [18] to find the hyper-parameters for the MLP model. Munoz et al. [132] used both simulated data from AIDA simulator and real data from D1namo dataset (CGM, insulin dosage, and CHO amount). In real scenario they used insulin data, though in D1namo dataset no insulin dosage has been recorded. They proposed separate input, LSTM, and Dense layers for meal intake, insulin dosages, and glucose data from CGM. Later on, the concatenated values of all those three outputs were fed through a dense layer and finally produced the prediction of BGLs.

Chen et al. [73] proposed a DL technique based on Dilated Recurrent Neural Network (DRNN) to predict BGL using the OhioT1DM dataset. Zhu et al. [140] extended the DRNN in [73] and gained a much larger receptive field in terms of neurons aiming at capturing long-term dependencies. They used transfer learning using data from multiple subjects to forecast future BGLs using the same OhioT1DM dataset. Wang et al. [137] presented stacked LSTM based DL models to forecast long-term BGLs taking only the BGL values as input.

Table 4: BGL Prediction Performance Comparison (**Real data**): Whole day (W), or Unknown (U); Metrics: RMSE (mg/dL) - on the 1st row, CEG (A+B) - on the 2nd row.

Ref	**Main Input Features**	**Dataset**	**Modeling Technique**	**Performances:** $\frac{RMSE}{CEG}$		
				PH=30	PH=60	PH=120
Physiological Models						
[40]	Glucose	12 T1DM, 10 days	Physiological Model	-	-	-
				98	96.5	-
[97]	Insulin, Meal	26 Patients, 3-week, T1DM and T2DM	Physiological Model	W:49.68	-	-
				W:92.5	-	-
[102]	Glucose, Insulin, Meal	10 adults, 2 weeks, T1DM	Physiological Minimal Model	17.67	30.36	40.46
				99.05	97.12	95.27
Traditional ML Models						
[43]	Glucose, Insulin, Exercise, Meal	15 T1DM, 5 to 22 days	Extreme Learning Machine	6.1	-	-
				-	-	-

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Table 4 – continued from previous page

Ref	Main Input Features	Dataset	Modeling Technique	Performances: $\frac{RMSE}{CEG}$		
				PH=30	PH=60	PH=120
[47]	Glucose	10 T1DM, 6 days	FNN, SOM, WFNN, LR	11.42	19.58	31
				93.84	89.34	82.37
[52]	Glucose, Insulin, Exercise, Meal	70 subjects, months [8]	Pre-clustered personalized linear regression model	U36.695	-	-
				-	-	-
[51]	Glucose	17 T1DM, 4 to 7 days	Non-linear auto-regressive NN	2.37	4.36	-
				-	-	-
[63]	Glucose, Insulin, Meal	10 T1D, 10 days	GE, RF	-	-	-
				98.18	94.76	91.19
[89]	Glucose	3 T1DM subjects, 3 days	Recursive least squares (RLS)	8.93	-	-
				99.36	-	-
[90]	Glucose	3 T1DM subjects, 60 hours	Kernel sparse filtering algorithms	17.57	-	-
				93.83	-	-
[80]	Glucose	12 T1DM, 14 days	SVR model based on Differential Evolution	10.78	12.95	-
				-	-	-
[82]	Glucose	9 T1DM patients, 12 days	ANN, 3 layers	17.4	-	-
				-	-	-
[87]	Glucose, Insulin, Exercise, Meal	6 T1DM, 8 weeks	SVR - Radial Basis Kernel	19.53	-	-
				-	-	-
[98]	Insulin, Meal, Heart Rate	32 T1D, 5-day	PBSID	26.33	-	-
				97.07	-	-
[109]	Glucose	25 T1D, 14 days, 5400 hr	RF	14.63	22.12	-
				-	-	-
[103]	Glucose	6 T1DM, 8 weeks	SVR, MLP	-	-	-
				98.76	-	-
[125]	Glucose	10 T1D, 4 days	PSO-KCCA	16.4	-	-
				-	-	-
[134]	Glucose, Insulin, Meal	Training: 16 patients, 30 days; Testing: 4 T1DM, 1 day	ARX, SVR; linear, bagging (bootstrap aggregating) and boost (LS-Boost) metaregressor	17.66	35.65	-
				92.70	71.79	-
[117]	Glucose	12 T1DM children, 7 days	MLP	6.31	15.33	-
				-	-	-
[120]	Glucose, Insulin	159 T1DM and T2DM patients	ANN	U2.3265	-	-
				-	-	-

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Table 4 – continued from previous page

Ref	Main Input Features	Dataset	Modeling Technique	Performances: $\frac{RMSE}{CEG}$		
				PH=30	PH=60	PH=120
[129]	Glucose	11 T1DM, over 7-50 nonconsecutive days, 162658 measurements	4 Kernel Ridge Regression (KRR) variants	20.48	-	-
				93.4	-	-
[118]	Glucose	5 T1DM, 6 days	Extreme Gradient Boosting	23.219	35.800	-
				-	-	-
[133]	Glucose	6 T1DM, 8 weeks	Fuzzy particle swarm optimization algorithm	27.46	-	-
				97.95	-	-
[133]	Glucose	89 T1DM, 26 hours	Fuzzy particle swarm optimization algorithm	38.93	-	-
				97.46	-	-
[122]	Glucose	6 T1DM, 8 weeks	Auto Regressive Time Delayed (ARTiDe) Jump NN	18.4	-	-
				-	-	-
[122]	Glucose	32 T1DM, 1 T2DM patients, 8 to 30 days (avg 15)	Auto Regressive Time Delayed (ARTiDe) Jump NN	18.7	-	-
				-	-	-
[143]	Glucose	124 T1D, 1240 days, 350,000 samples	30 ML algorithms, FNN	21.52	-	-
				-	-	-
[88]	Glucose	100 T1DM, T2DM, 4 Days	ARIMA	RAD:5.4	TG(30):.48	
				SSGPE:7.21		-
[123]	Glucose, Insulin, Meal	112 patients, T1DM, over 90 days, 1639921 CGM values	RF w/ VIP	se:93.61	91.01	-
				sp:93.5	89.82	-
Deep Learning Models						
[66]	Glucose, Insulin, Meal	10 T1DM patients, 400 days	LSTM	21.4	38.0	-
				-	-	-
[65]	Glucose	25 children, T1DM, 160 readings/subject	Semi Supervised Deep Learning	21.4	38.0	-
				93.23	-	-
[83]	Glucose	20 T1DM Adults, 1 week, Readings: (6975 ± 1612)	LSTM, BiLSTM	21.747	36.918	-
				-	-	-
[73]	Glucose, Insulin, Exercise, Meal	6 T1DM, 8 Weeks	Dilated RNN	19.04	-	-
				-	-	-
[100]	Glucose, Insulin, Exercise, Meal	10 T1DM	CNN	21.07	33.27	-

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Table 4 – continued from previous page

Ref	Main Input Features	Dataset	Modeling Technique	Performances: $\frac{RMSE}{CEG}$		
				PH=30	PH=60	PH=120
				-	-	-
[104]	Glucose, Insulin, Meal	six T1DM patients, eight weeks	LSTM	17.99	28.20	-
				-	97.31	-
[101]	Glucose, Insulin, Meal	10 T1DM, 180 days	Dilated CNN	19.19	31.78	-
				-	-	-
[101]	Glucose, Insulin, Exercise, Meal	6 T1DM, 8 weeks	Dilated CNN	19.28	31.83	-
				-	-	-
[94]	Glucose, Exercise, Meal	10 T2DM Patients, 6 months	LSTM, Transfer learning	-	-	-
				W99.34	-	-
[132]	Glucose, Insulin, Meal	9 TIDM patients, 4 days	Stacked LSTM	6.42	11.35	-
				98.08	87.22	-
[124]	Glucose, Insulin, Meal	34 T1DM subjects, 102 to 268 days	CNN, LSTM	9.18	-	-
				-	-	-
[119]	Glucose, Insulin, Exercise, Meal	6 T1DM patients, 8 weeks, 2018	LSTM	10.66	25.183	-
				100	98	-
[137]	Glucose	20 T1DM, 878k Data points	Stacked LSTM	11.96	19.24	-
				97.6	95	-
[140]	Glucose, Insulin, Exercise, Meal	6 T1DM, 8 Weeks	Dilated RNN	18.9	-	-
				-	-	-
[139]	Glucose, Insulin, Exercise, Meal	six T1DM patients, eight weeks	ARX, LSTM, TCN, 7 regression algorithms	W19.48 (ARX,Ridge)	-	-
				-	-	-
[128]	Glucose, Insulin, Meal	24 T1DM patients, 16 CSII, 8 MDI, 28+2+2 days	Physiologically-informed FNN	-	28	38
				-	97	94
[127]	Glucose	20 T2DM, 3-7 days	RNN (GRU)	21.46	-	-
				99.03	-	-
[131]	Glucose	50 T1DM patients, 14 days	LSTM, ARIMA	20.8026	36.5706	-
				-	-	-
[78]	Glucose	40 T1DM patients, 3 years, 550k GL measurements	RNN Autoencoder	MAPE:4.59		-
				-	-	-

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Table 4 – continued from previous page

Ref	Main Input Features	Dataset	Modeling Technique	Performances: $\frac{RMSE}{CEG}$		
				PH=30	PH=60	PH=120
Hybrid: Physiological + Traditional ML Models						
[28]	Glucose, Insulin, Exercise, Meal	15 Patients, T1DM, 5-22 days	Physiological Minimal Model and SVR Model using Gausssian RBF kernel	6.03	7.14	7.62
				97.51	96.81	97.48
[31]	Glucose, Insulin, Meal	15 T1DM patients, 7 days	Linear model, Meal Absorption Model, FNN	14.0	-	-
				-	-	-
[36]	Glucose, Meal	20 T1DM patients, 2 or 3 consecutive days	Linear model, Meal Absorption Model, FNN, Skip connection	16.6	-	-
				-	-	-
[33]	Glucose, Insulin, Exercise, Meal	5 T1DM, 200 timestamps, 40 points per patient	Physiological + SVR	19.5	35.7	-
				-	-	-
[70]	Glucose, Insulin, Exercise, Meal	6 T1DM, 8 weeks	Physiological features and ANN	19.33	31.72	-
				-	-	-
[126]	Glucose, Insulin, Meal	5 patients (4-T2DM, 1-T1DM), 3 weeks	Physiological Minimal Model and FNN	-	20.16	31.5
				96.46	96.03	-
[142]	Glucose, Insulin, Meal	5 patients (4-T2DM, 1-T1DM), 3 weeks	Physiological Minimal Model and ANN	-	26.75	30.92
				-	-	-
Hybrid: Physiological + Deep Learning Models						
[81]	Glucose	6 patients, 8 weeks, T1DM, 2018	LSTM + physiological loss function	20.1	33.2	-
				-	-	-
[130]	Glucose	6 patients, 8 weeks, T1DM, 2018	LSTM + physiological loss function, Noise Experiments	18.867	31.403	-
				-	-	-

5.3 Hybrid Models

Hybrid models is usually a combination of two or more processes. In most of the cases, this model is a two step process. In the first phase or step, physiological models are designed to represent CHO digestion, glucose absorption, insulin absorption, exercise model, etc. In the second phase, the output from the physiological modes is provided as input to a data-driven model, which eventually predicts the BGL.

Georga et al. [28] used compartmental models of the glucose-insulin regulatory system [5, 7, 16] to generate CHO absorption rate related input features. Then, they used those features along with CGM values, insulin dosage, exercise, and demographic information as input to a patient-specific SVR [14] based predictive model for learning the glucose metabolism and for predicting BG concentration.

Contreras et al. [60] applied a hybrid approach on simulated data. They built physiological models (Insulin on board, CHO absorption) for insulin and meal. Then they take these models as input and used symbolic regression through Grammatical Evolution (GE) [39] to determine the underlying glucose dynamics. Grammatical evolution is a search algorithm that uses a grammar consisting of a set of rules to build expression for BG considering previous BGLs, CHO intake, and insulin dosage. These are used to generate predictive time series models according to a predefined objective function. Contreras et al. used a penalizing fitness function as objective function to stop deviations from the target BGL.

Martinsson et al. [81] proposed a RNN with a glucose specific physiologically motivated loss function for predicting BGLs along with an estimation of its certainty using only the BGLs from the OhioT1DM dataset as input. The loss function combines the Mean Squared Error (MSE) with a penalty term for predictions that would lead to clinically dangerous treatments. The output vector from the final LSTM cell is fed through a FC output layer having two outputs with a linear activation function. The output is modeled as a univariate Gaussian distribution, which gives an estimate of the confidence in the models' predictions. Martinsson et al. [130] used LSTMs to predict BGL. Similar to [81], their model provided an estimate of the certainty of the predicted value by modeling it as a univariate Gaussian distribution. They used a physiological loss function as contraindicated interventions.

Karim et al. [126] used a two compartmental physiological model proposed in [11], which consists of stomach compartment, gastric emptying, intestinal compartment, breakdown of starch, and intestinal glucose absorption. This model takes the macro-nutrient content of every logged meal as input and provides the glucose absorption curve. Then they provided the absorption curve along with the insulin dosage and BGLs as input to a FNN with 20 hidden layers. In their latest work [142], they added two additional parameters which are total area under the absorption curve, and basal dosage time. This improved their 120 minutes RMSE to 30.92.

Zecchin et al. [31] used a linear prediction algorithm proposed in [20] with past CGM values as input to produce the glucose prediction model. They used the physiological meal simulation model proposed in [19] to get the glucose rate of appearance. Along with these two, other 6 inputs including insulin dosages were fed to a fully connected FNN that has one hidden layer with eight neurons, each one with a tangent sigmoid activation function and an output layer. The final output comes by combining the linear glucose prediction model and the prediction error from the NN. Zecchin et al. [36] used the same model in [31] with an additional skip connection (the jump part) from input of NN to the output.

Saiti et al. [134] used an open compartment model [15] to model the impact of bolus insulin absorption. Along with this physiological model all other input data: BGLs and CHO amounts were fed to ARX and SVR as individual prediction models. Then, they implemented and compared three different ensemble methods: linear, bagging (bootstrap aggregating) and boost (LS-Boost) metaregressor, for combining results from individual ARX and SVR models for final prediction. Bunescu et al. [33] used a physiological model of blood glucose dynamics based on [23], with a few adaptations in order to better match their published data and feedback from their doctors. The physiological model generated informative features for a SVR model that is trained on patient specific data. They used CGM readings, insulin dosages, exercise, CHO amounts, and sleep time as their input data.

Table 5: BGL Prediction Performance Comparison (**Simulated data**): Whole day (W), or Unknown (U); Metrics: RMSE (mg/dL) - on the 1st row, CEG (A+B) - on the 2nd row.

Ref	**Main Input Features**	**Dataset**	**Modeling Technique**	**Performances:** $\frac{RMSE}{CEG}$		
				PH=30	PH=60	PH=120
Physiological Models						
[40]	Glucose	10 adults, 4 days	Physiological Model	- 98.5	- 97.25	- 88

Continued on next page

Table 5 – continued from previous page

Ref	Main Input Features	Dataset	Modeling Technique	Performances: $\frac{RMSE}{CEG}$		
				PH=30	PH=60	PH=120
[102]	Glucose, Insulin, Meal	10 subjects, T1D, 2 weeks	Physiological Minimal Model	10.06	22.56	34.74
				99.62	97.15	94.16
Traditional ML Techniques						
[59]	Glucose, Insulin, Meal	300 BG profile, 2100 sample data, 9 hours	Two-layer NN model with 8 hidden neurons	5.9	-	-
				-	-	-
[89]	Glucose	10 Adults, 3 days	Recursive Least Squares (RLS) and 3 variants	9.86	-	-
				99.65	-	-
[90]	Glucose	10 Adults, 6 days	Kernel sparse filtering algorithms	13.74	-	-
				91.96	-	-
[55]	Glucose	5 T1DM subjects, 25 days	SVR	AMRE:1	9.8%	-
				-	-	-
Deep Learning Models						
[104]	Glucose, Insulin, Meal	40 patients, 600 days	LSTM	1.23	2.27	-
				-	-	-
[104]	Glucose, Insulin, Meal	10 T1D, 90 days	LSTM	2.93	4.92	-
				-	99.47	-
[101]	Glucose, Insulin, Meal	10 T1D, 180 days	Dilated CNN	8.8	19.9	-
				-	-	-
[100]	Glucose, Insulin, Exercise, Meal	10 T1D, 360 days	CRNN	9.38	18.87	-
				-	-	-
[140]	Glucose, Insulin, Meal	10 T1D, 360 days	Dilated Recurrent Neural Network (DRNN)	7.8	-	-
				-	-	-
[116]	Glucose, Insulin, Meal	100 T1D, four-day + single T1D real patient, a month	Stacked LSTM	U21.09	-	-
				-	-	-
[132]	Glucose, Insulin, Meal	40 T1D, 10 days	Stacked LSTM	2.83	4.35	-
				99.86	99.86	-
Hybrid: Physiological + Traditional ML						
[31]	Glucose, Insulin, Meal	20 T1D, 11 days	Linear model, Meal Absorption Model, FNN	9.4	-	-
				-	-	-
Hybrid: Others						
[60]	Glucose, Insulin, Meal	100 subjects, 14 days	Hybrid: Physiological + Grammatical Evolution	M20.96	-	-
				M98.31	-	-

5.4 Comparative Analysis

Various modeling techniques have been used to predict BGL. These are being categorized in Table 3. Among them, DL techniques (38%) and traditional ML techniques (43%) as seen in Figure 7 have been used majority of the times followed by hybrid models (13%) and physiological models (6%). Among the traditional ML techniques, FNN is the most popular followed by SVR and Auto Regressive (AR) models. Among the DL techniques, LSTM models are the most popular followed by RNN and CNN.

Physiological models have been used by the research community the least due to the need of expert knowledge, the lack of patient personalizing capabilities, and the inability to consider all the physiological mechanisms. But, the research community is more willing to use them as a data pre-processing step of the data-driven models.

Following the current trend in other areas, the researchers are using data driven modeling techniques in the area of diabetes research. This is due to the recent i) availability of plethora of data from Internet of Thing (IOT) devices like CGM devices, Insulin pumps, Wearable fit bands, EHRs, etc, and ii) advent of high performing Graphical Processing Units (GPUs) to run the data driven techniques processing a lot of data in short period of time.

5.5 Performance Measure Analysis

While reviewing different research works, articles relevant to BGL prediction are extensively studied. All the articles that are studied are listed in Table 4 and Table 5. There are a lot of variations with respect to diabetes type, data set type, data set size, input features, prediction horizon, and performance metrics. These variations make it a challenging task to do a fair comparison among the predictive performances of different modeling techniques. A modeling technique with best performance with a proprietary dataset may have lower performance with other datasets than other modeling techniques. But, few intriguing insights can be taken away for future improvement.

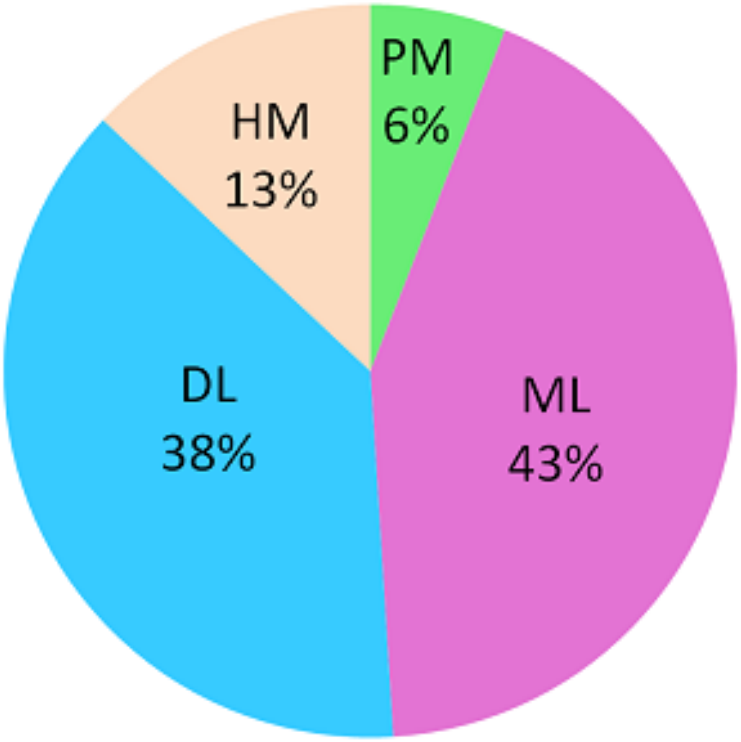


Figure 7: Categories of different modeling techniques.

We used the two most widely used performance metrics in diabetes research: Root Mean Squared Error (RMSE) and Clarke Error Grid (CEG). RMSE quantifies the statistical accuracy of a modeling technique. Lower RMSE values are better. Whereas, CEG quantifies the clinical accuracy of a modeling technique. It has five regions: A, B, C, D, and E. Higher CEG values for A and B are

better. Various types of datasets including clinical, simulated, EHRs, and pathological test reports are being used. Among the reviewed articles, in 72.4% of the cases only clinical datasets were used. In 22.4% of the cases both clinical and simulated datasets were used. Due to high rate of data generation by CGM devices per day, 96.5% of the clinical datasets have T1DM data. Most of those datasets are proprietary. Few datasets: OhioT1DM, D1namo, Pima Indian Data, UCI, DIAdvisor, DirecNet, Tidepool are publicly available. Among those, OhioT1DM dataset is the most popular [70, 73, 81, 87, 101, 104, 119, 122, 124, 130, 133, 139, 140] and is used 24.1% of the times. It would be a fair comparison of the performances of the research works that used the OhioT1DM dataset. It is observed that the best performance is achieved in the research work done by [124] with RMSE 9.18. They applied CNN and LSTM on previous glucose history, insulin dosage, and meal input data. The second best performance is achieved in the research work of [119] with RMSE 10.66. They also applied LSTM on previous glucose history, insulin dosage, physical activity, and meal. Performances of the rest of the works are close. Majority used DL e.g., LSTM, RNN, CNN. Few used traditional ML techniques such as ARX, NN, and SVR.

In 25.86% of the reviewed articles simulated data were used. Those data were generated from two simulators: UVA/Padova and AIDA. It is observed that, the performances of the modeling techniques applied on simulated data [Table 5] is considerably better than when applied on real data [Table 4] across the board.

Input data used by reviewed articles vary. Mostly, they are various combinations of the four critical input features: previous glucose readings, insulin dosages, physical activity, and meal intake. Among the research articles that used physiological models [Table 4], we can see that Bock et al. [40] used only previous glucose readings. Their model's prediction performance is 98% in Clerke Error Grid (CEG) for PH 30. Gyuk et al. [97] performed worse without CGM reading. Liu et al. [102] performed better when they used more information in addition to previous CGM readings.

From the performances of the proposed algorithms in Table 4, it is observed that all the high performing modeling techniques with respect to statistical metric RMSE values used previous glucose readings. Majority of them only used previous glucose readings. The best performing article by [120] used ANN on glucose readings and insulin dosage. The second best performing article by [51] used Non-linear auto-regressive NN on glucose readings only. Among the articles that provided both statistical metric RMSE and clinical metric CEG, the best performance is observed by [28]. They used SVR Model using Gaussian RBF kernel on all four input features. Though their 30 and 60 minutes CEG is not that good at 97.51 and 96.81 their 2-hr CEG has the best value at 97.48. The best performance in CEG is observed by [119]. They showed CEG values of 100 at 30 minute and 98 at 60 minute. They had reasonable performance in RMSE with 10.66 at 30 minute and 25 at 60 minute. They used LSTM on all four input features.

Ozogur et al. [133] used Fuzzy particle swarm optimization algorithm on a large cohort and received a 38.93 RMSE on its prediction performance. When they used a smaller cohort they received a better performance. For data-driven models, with more data prediction performance gets better. Here, it is observed that with Fuzzy algorithm the prediction performances is better on smaller datasets. Whereas, DAntoni et al. [122] got similar performances when they used Auto Regressive Time Delayed (ARTiDe) Jump Neural Network (Jump NN) on a cohort of 33 patients for on average 15 days and on a different smaller cohort of 6 patients for 8 weeks. Mirshekarian et al. [104] achieved a performance improvement when they used memory-augmented MemLSTM neural attention model over their previous LSTM model [66]. This gives an observation that attention aware models generally improves BGL prediction performance.

This study recorded prediction performances for only three Prediction Horizons (PH): 30 minutes, 60 minutes, and 120 minutes. Among them PH 30 minutes is the most widely used. Whereas 120 minutes is the least popular. From Table 4 it is observed that, in general, increasing prediction horizon decreases performance. That means the prediction performance of BGL in 2-hrs from current time is usually worse than in 30 minute. This is due to the complexity of glucose-insulin dynamics. It involves quite a few factors. With longer prediction horizon, it gets more challenging to predict the dynamics among these factors. Hence, prediction performance suffers. BGL two hours after meal is a standard metric used to check whether BGL is within the normal range or not. More researches should provide

BGL prediction for PH 120 minutes. Among the BGL prediction performance metrics, RMSE is the most widely used followed by CEG.

6 Conclusion

In this paper, we have conducted a comprehensive literature review on the latest efforts on diabetes management as a part of clinical applications. We have analyzed the existing recent works from different perspectives that include clinical applications, input features, modeling techniques, and the performances. At first, we have presented the detail study, and designed the taxonomy from the perspective of different clinical applications. Among the clinical applications, BGL prediction is found to be the most popular among the researchers. Then, we have discussed the details of different input features such as meal intake, historical glucose level, etc. used in our focused BGL prediction application. We have observed lack of research on T2DM though over 90% of diabetic patients have T2DM. Due to high data point rate from CGM devices, majority of the research works used T1DM data. There are few publicly available T1DM datasets including OhioT1DM. Moreover to the best of our knowledge, there is no publicly available T2DM dataset. T2DM data collection is challenging as it takes much longer data collection period. There is a dire need of publicly available benchmark T2DM dataset. More research efforts are needed to be directed to collect data from T2DM patients. Now-a-days lots of data are generated from wearable devices. Therefore, more efforts are needed to be directed to identify the type and intensity of activities from wearable data. Self diabetes management is also becoming the new trend as it empowers a person to manage his/her diabetes. In this direction, instantaneous prediction of BGL from meal image is required, which can be useful research direction.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

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